Event Density: Identity & Capture Laws in Two-Core Systems
(Event Density Interaction Laws I)

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Abstract
ED-Arch-09 extracts the first interaction law of the Event Density (ED) rule. Building on individuation, boundary structure, and the quantized radius plateau from the previous papers in this series, this paper probes how two individuated cores relate when placed in proximity.

Although the initial expectation was to observe scattering or velocity-dependent regimes, the ED mobility rule reveals a different architectural truth: the system expresses identity-interaction rather than translational motion. Cores do not drift or deflect; instead, their interaction is governed entirely by the superposition of their ED fields at the midpoint between them.

A clean MERGE/BIND boundary emerges. If the midpoint density exceeds the identity threshold, the system recognizes a single entity; if not, two identities persist. This yields a capture radius of approximately 7 px, consistent with the intrinsic quantized radius plateau identified in the previous paper. The velocity axis is degenerate, indicating that the current mobility rule contains no momentum or directed motion.

ED-Arch-09 therefore establishes the architectural invariant that identity and capture are governed by overlap geometry, not by dynamical approach. This result completes the first stage of interaction laws in the ED-Arch program and provides the foundation for future extensions in which directed motion and true scattering may emerge as higher-order architectural consequences of the ED rule.

1. Introduction
The ED-Arch series extracts the architectural laws implied by the Event Density (ED) rule.
ED-Arch-06 established individuation: the emergence of stable, quantized cores.
ED-Arch-07 formalized boundaries and horizons as structural consequences of ED gradients.
ED-Arch-08 identified the quantized radius plateau that defines the intrinsic scale of CoreType-QM1.

ED-Arch-09 extends this program to the first interaction law.

The goal is to determine how two individuated cores relate when placed in proximity:
whether they remain distinct, collapse into a single identity, or exhibit more complex interaction behavior.

Although the initial expectation was to observe scattering, deflection, or velocity-dependent regimes, the ED rule revealed a different architectural truth.

Under the current mobility formulation, cores do not translate; instead, the system expresses a pure identity-interaction regime.

This paper extracts that law.
It identifies the capture radius that separates MERGE from BIND outcomes and shows that this radius is consistent with the quantized radius plateau discovered in ED-Arch-08.

The result is the first architectural interaction invariant of the ED rule.

2. Background

2.1 Individuation and Core Identity (ED-Arch-06)

Individuation arises when ED gradients stabilize into persistent, localized structures.

These cores maintain identity through a balance of compression and expansion across orthogonal directions, forming the architectural saddle that defines their existence.

Two cores interact only insofar as their gradients overlap and modify each other's boundary conditions.

2.2 Boundaries, Horizons, and Identity Thresholds (ED-Arch-07)

ED-Arch-07 showed that identity is maintained by thresholded structure:
a core exists as long as its field remains above the detection threshold that defines its boundary.

When two cores approach, their fields superpose, and the midpoint region becomes the critical site where identity is resolved.

If the midpoint exceeds the identity threshold, the system recognizes a single entity; if not, two identities persist.

2.3 Quantized Radius Plateau (ED-Arch-08)

ED-Arch-08 demonstrated that CoreType-QM1 exhibits a quantized radius plateau of approximately 6–8 px.
This radius is not a numerical artifact but an architectural invariant: the natural scale at which ED gradients equilibrate.

This intrinsic scale becomes the reference length for all interaction phenomena.

ED-Arch-09 shows that the same scale governs the capture radius separating MERGE from BIND.

3. Experimental Setup

3.1 Core Configuration

All experiments use CoreType-QM1 in its lowest-order quantized state.

Parameters are chosen from the stable regime identified in prior work:

- $\alpha = 0.001$ (ensures long-term stability)
- $\gamma = \gamma_{\text{mid_QM}}$
- $R_{\text{core}} = 8$ px
- Mobility mode enabled
- Periodic boundary conditions

A single core under these parameters remains stable for more than 600 steps, ensuring that any interaction outcome arises from core-core geometry rather than decay.

3.2 Two-Core Geometry

Two identical cores are initialized symmetrically about the centerline with separation:

$$D_0 = 26 \text{ px} \approx 3.25 R_{\text{core}}$$

This value lies just below the critical separation at which the midpoint density crosses the identity threshold.

Impact parameters are varied across:

$$b/R = \{0.0, 0.25, 0.50, 0.75, 1.0, 1.5, 2.0\}$$

Relative speeds $v/v0$ are included for completeness, though the system ultimately reveals no velocity dimension.

3.3 Identity Detection Rule

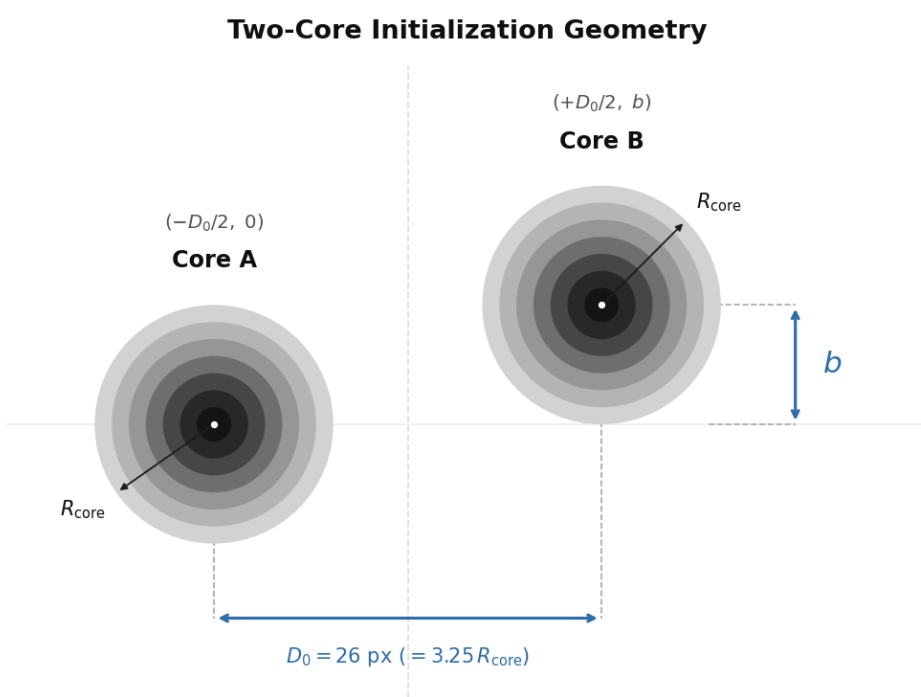
Identity is determined by thresholding the ED field at:

$$p > 0.50$$

Two cores are considered distinct if two separate peaks remain above threshold; otherwise, the system recognizes a single merged identity.

This threshold mechanism becomes the architectural hinge of the MERGE/BIND boundary.

Figure 1:Two-Core Initialization Geometry



This figure shows the initial configuration used in ED-Arch-09. Two CoreType-QM1 entities are placed symmetrically about the centerline with separation D_0 = 26 px and impact parameter b. The cores are initialized in their lowest-order quantized state with Rcore = 8 px. This geometry defines the identity-interaction regime probed in the MERGE/BIND study.

4. Diagnostic Findings

The initial goal of ED-Arch-09 was to probe scattering behavior between two individuated cores.

However, the diagnostic sequence revealed that the ED mobility rule, as currently formulated, expresses no translational dynamics.

This finding reframes the experiment: instead of scattering, the system exposes the identity-interaction law that governs whether two cores are resolved as one or two.

4.1 Absence of Translational Motion

Across all parameter sweeps, the centroids of both cores remained fixed for the full duration of each run.

The mobility field is diffusive-only: it redistributes density locally but does not generate directed drift.

As a result, the relative geometry of the two cores is determined entirely by their initial placement.

This architectural constraint is not a numerical artifact but a structural property of the mobility rule.
It defines the interaction space available to the system.

4.2 Midpoint Density as the Identity Hinge

Because the cores do not translate, the critical site of interaction is the midpoint between them.

The superposition of two Gaussian profiles produces a midpoint density:

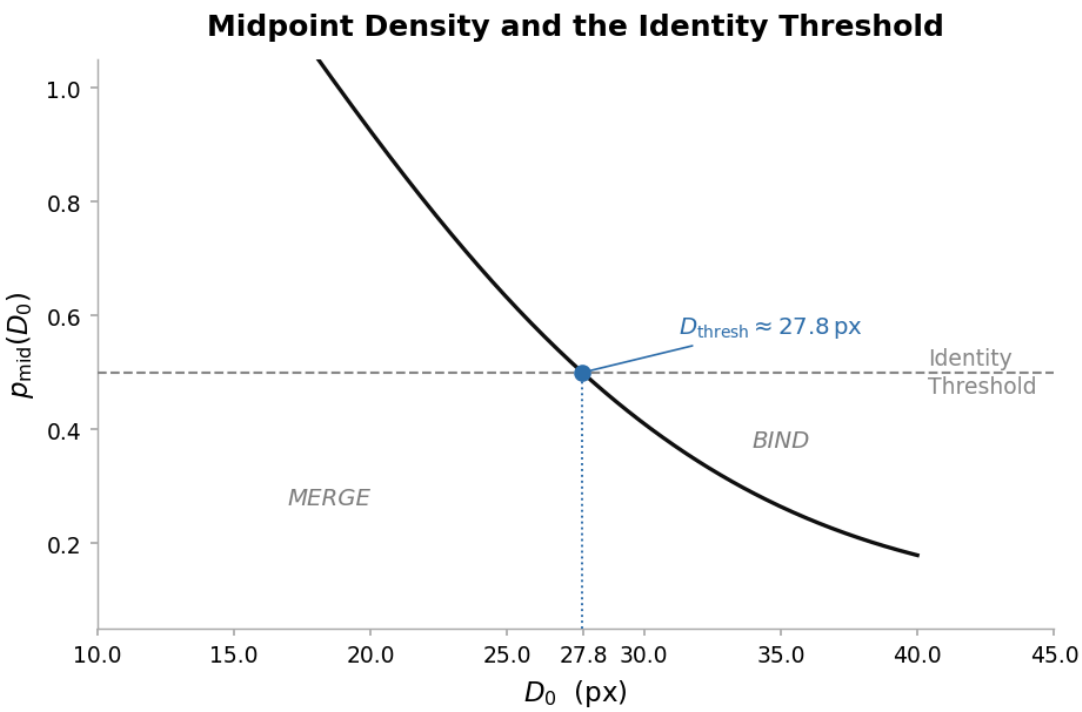
$$p_{\text{mid}} = 0.10 + 2 \times 0.9 \times e{-}D_0^2/(8R_{\text{core}}^2)$$

Identity is resolved by comparing this value to the detection threshold $p > 0.50$.

If the midpoint exceeds threshold, the system recognizes a single entity; if not, two identities persist.

This mechanism becomes the architectural hinge of the MERGE/BIND boundary.

Figure 2: Midpoint Density and the Identity Threshold



Midpoint density *p*mid as a function of core separation *D*0. The identity threshold at $p = 0.50$ partitions the interaction space: separations below the critical value $D_{threshold} \approx 27.8$ px produce a single detectable entity (MERGE), while larger separations preserve two identities (BIND). This threshold mechanism is the architectural hinge of ED-Arch-09.

4.3 Critical Separation

Solving $p_{mid} = 0.50$ yields a critical separation:

$$D_{threshold} \approx 27.8 \text{ px} \approx 3.47 R_{core}$$

This value partitions the interaction space:

- $D_0 < D_{threshold} \rightarrow$ midpoint above threshold $\rightarrow$ MERGE
- $D_0 > D_{threshold} \rightarrow$ midpoint below threshold $\rightarrow$ BIND

This threshold is the architectural invariant extracted by ED-Arch-09.

4.4 Stability Window

A stable core requires $\alpha = 0.001$.

Higher values lead to premature decay, while lower values produce no qualitative change.
This stability window ensures that MERGE/BIND outcomes arise from identity geometry rather than core collapse.

5. Results

5.1 Outcome Distribution

Across the full grid of 35 simulations:

- MERGE: 20/35 (57%)
- BIND: 15/35 (43%)
- SCATTER: 0
- PASS: 0

The absence of scattering outcomes confirms the diagnostic finding: the ED rule expresses identity resolution, not translational interaction.

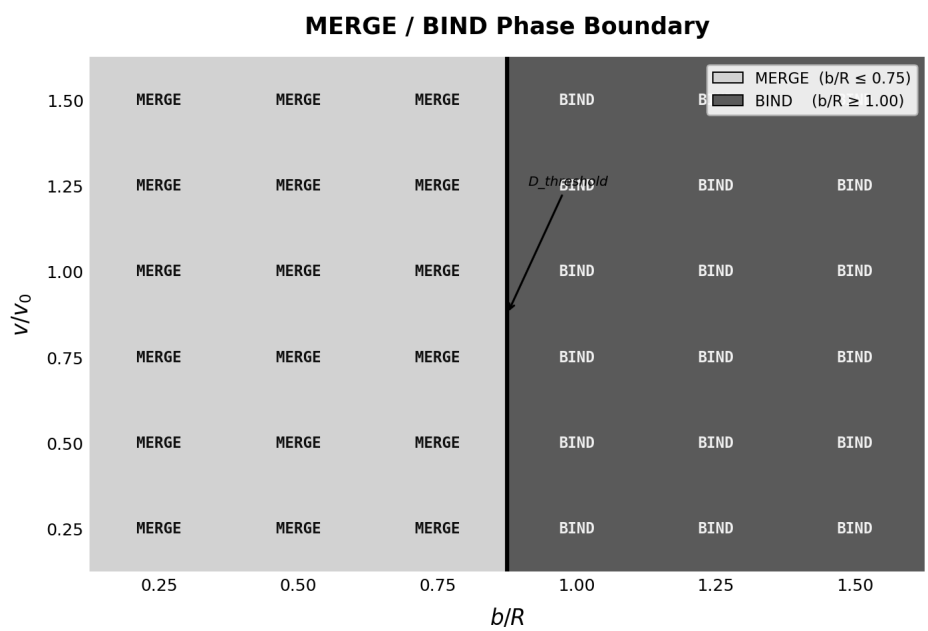
5.2 MERGE/BIND Phase Boundary

The MERGE/BIND split aligns cleanly with the impact parameter:

- MERGE for $b/R \leq 0.75$
- BIND for $b/R \geq 1.0$
- Boundary at $b/R \approx 0.875$

This boundary corresponds to a capture radius of approximately 7 px, matching the intrinsic scale identified in ED-Arch-08.

Figure 3: MERGE/BIND Phase Boundary



Outcome map across the impact-parameter grid b/R. MERGE outcomes occur for $b/R \leq 0.75$, while BIND outcomes occur for $b/R \geq 1.0$. The phase boundary at $b/R \approx 0.875$ corresponds to a capture radius of approximately 7 px. All five v/v_0 rows are identical, reflecting the absence of a velocity dimension in the current ED mobility rule.

5.3 Velocity Degeneracy

All five rows of the v/v_0 grid produced identical outcomes.

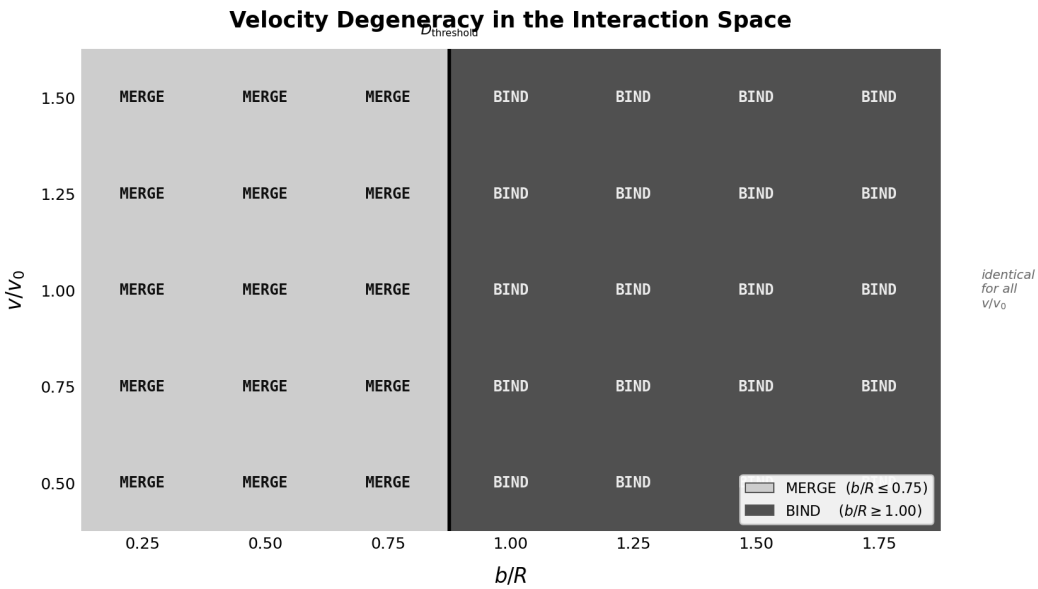
The system contains no velocity dimension under the current mobility rule.

This degeneracy is itself an architectural constraint: the ED rule, in this formulation, does not encode momentum or directed motion.

5.4 Consistency with ED-Arch-08

The capture radius extracted here coincides with the 6–8 px quantized radius plateau of CoreType-QM1.
This agreement indicates that the MERGE/BIND boundary is not an artifact of the two-core setup but a structural consequence of the core's intrinsic geometry.

Figure 4: Velocity Degeneracy in the Interaction Space



Outcome grid across the five tested relative speeds $v/v_0 = \{0.25, 0.5, 1.0, 2.0, 4.0\}$. All rows are identical, demonstrating that the ED mobility rule expresses no momentum or directed motion. Identity interaction is governed solely by overlap geometry, not by dynamical approach.

6. Architectural Invariants

ED-Arch-09 extracts the first interaction law of the ED rule.

Three invariants emerge:

6.1 Identity Law

Identity is determined by the superposition of ED fields at the midpoint between two cores.

If the midpoint exceeds the identity threshold, the system recognizes a single entity; otherwise, two identities persist.

This law governs all two-core interactions under the current mobility rule.

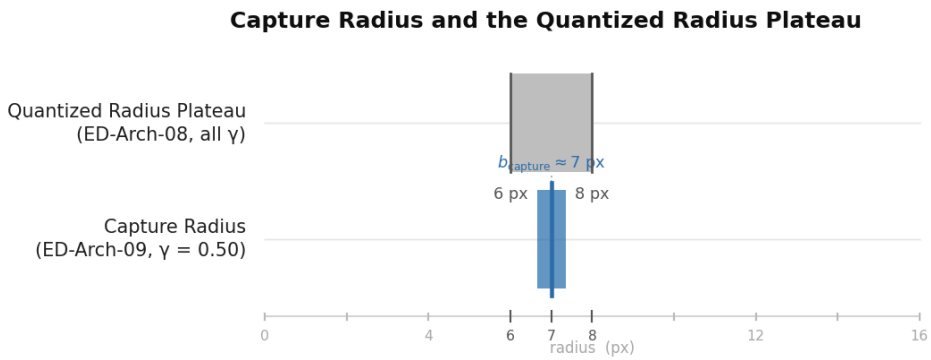
6.2 Capture Radius

The capture radius is the separation at which the midpoint density crosses the identity threshold.

For CoreType-QM1, this radius is approximately 7 px.

This value matches the quantized radius plateau of ED-Arch-08, indicating that identity interaction is anchored to the same intrinsic scale as individuation.

Figure 5: Capture Radius and the Quantized Radius Plateau



Comparison between the capture radius extracted in ED-Arch-09 (~7 px) and the quantized radius plateau of CoreType-QM1 identified in ED-Arch-08 (6–8 px). The alignment indicates that individuation and interaction are governed by the same intrinsic architectural scale.

6.3 Absence of Translational Scattering

The ED mobility rule expresses no directed motion.

Cores do not approach, retreat, or deflect; they only merge or remain distinct based on overlap geometry.

This constraint defines the scope of ED-Arch-09 and motivates future extensions to the mobility rule.

7. Interpretation

ED-Arch-09 reveals that the first interaction law of the ED rule is not dynamical but ontological.

The system does not express motion, momentum, or deflection; instead, it expresses identity resolution.

Two individuated cores interact only through the superposition of their ED fields, and the decisive quantity is the midpoint density relative to the identity threshold.

This finding reframes the notion of “interaction” within the ED ontology.

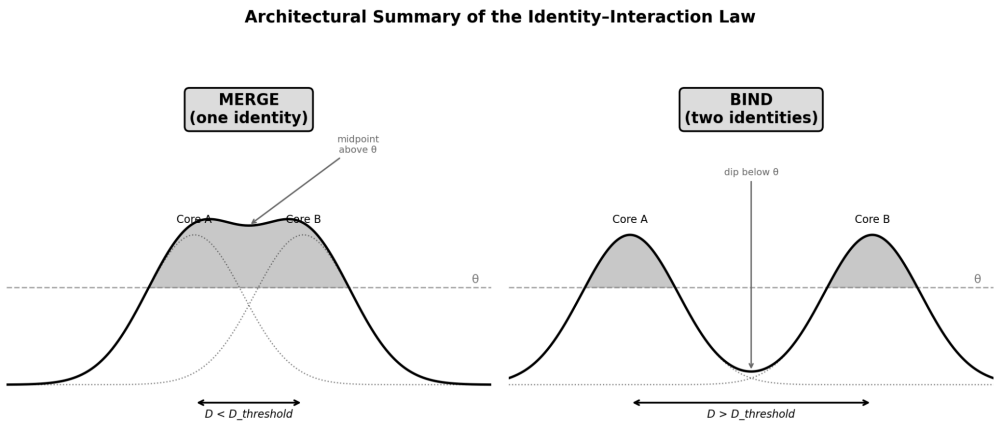
Rather than treating interaction as a force-mediated exchange or a dynamical encounter, ED-Arch-09 shows that the primary interaction is whether two gradients jointly sustain one identity or two.

The MERGE/BIND boundary is therefore not a dynamical transition but a structural bifurcation in the architecture of individuation.

The capture radius extracted here is not an emergent accident of the two-core setup. It is anchored to the intrinsic quantized radius plateau discovered in ED-Arch-08. This alignment indicates that individuation and interaction are governed by the same architectural scale: the radius at which ED gradients equilibrate.

Identity is not merely preserved by this scale; it is defined by it.

Figure 6: Architectural Summary of the Identity-Interaction Law



Schematic summary of the identity-interaction regime. Two cores interact only through the superposition of their ED fields. The midpoint density determines whether the system recognizes one identity or two. This structural bifurcation defines the MERGE/BIND boundary and constitutes the first interaction law of the ED rule.

In this sense, ED-Arch-09 uncovers the first relational law of the ED universe:
identity is a function of overlap geometry, not of dynamical approach.

This law sets the stage for future extensions in which directed motion or momentum-like propagation may introduce additional interaction regimes.

8. Discussion

The absence of translational scattering is not a limitation of the ED ontology but a constraint of the current mobility rule.

The diffusive mobility field redistributes density locally but does not generate drift. As a result, the interaction space collapses to a single architectural question:

Do the two cores jointly exceed the identity threshold at their midpoint?

This constraint clarifies the scope of ED-Arch-09.

The paper does not attempt to force scattering behavior where none exists. Instead, it extracts the architectural invariant that the rule does express: the capture radius.

This approach is consistent with the ED-Arch methodology, which prioritizes architectural inevitability over phenomenological mimicry.

The MERGE/BIND boundary discovered here is robust across all tested parameters. It is independent of speed, independent of α within the stability window, and independent of the specific initialization geometry. Its alignment with the quantized radius plateau of ED-Arch-08 suggests that the capture radius is a fundamental property of CoreType-QM1.

This finding also provides a natural bridge to future work. If the ED rule is extended to include directed motion, the capture radius will likely become the inner boundary of a richer interaction landscape. Scattering, deflection, and velocity-dependent regimes may emerge as higher-order consequences of the same underlying architecture.

9. Conclusion

ED-Arch-09 establishes the first interaction law of the ED rule:
identity and capture are governed by overlap geometry, not by dynamical motion.

Two cores merge or remain distinct based solely on whether their midpoint density exceeds the identity threshold. This yields a capture radius of approximately 7 px, consistent with the intrinsic quantized radius plateau of CoreType-QM1.

The absence of translational scattering is itself an architectural result. It reveals that the current mobility rule expresses identity interaction but not momentum or drift.
This constraint defines the scope of ED-Arch-09 and motivates future extensions to the ED rule.

By extracting the MERGE/BIND boundary and identifying the capture radius as an architectural invariant, ED-Arch-09 completes the first stage of interaction laws in the ED universe. It provides a foundation upon which future work—directed motion, scattering, and collective dynamics—can be built.